

MINOR PROJECT 2

Credit Card Customer Segmentation Using K-Means Clustering

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Project	Credit Card Customer Segmentation

1. Title

Credit Card Customer Segmentation Using K-Means Clustering

2. Problem Statement

Financial institutions manage thousands of customer transactions every day. This project uses KMeans Clustering to group customers with similar purchasing behaviour, payment history, credit utilization and cash advance activities.

3. Dataset Description

Dataset: CC GENERAL.csv (Kaggle). Total Records: 8950. Total Attributes: 18. Target Variable: None. The dataset contains balance, purchases, cash advance, payments, credit limit and tenure information.

4. Data Preprocessing Steps

Loaded dataset, checked missing values, filled missing values using mean, removed duplicate records, dropped CUST_ID, standardized all numerical features using StandardScaler.

5. Exploratory Data Analysis

Performed statistical summary, feature distribution histograms, correlation heatmap, boxplots, pairplot and PCA visualization to understand customer behaviour and feature relationships.

6. Model Used

K-Means Clustering. The Elbow Method was used to select the optimal number of clusters.

7. Training Process

Standardized features, selected K=4 using the Elbow Method, trained the K-Means model, assigned cluster labels and visualized clusters using PCA.

8. Evaluation Metrics and Results

Silhouette Score: 0.198.

Four customer segments were identified: Regular Customers, High-Value Customers, CashAdvance Dominant Customers and Low-Activity Customers.

9. Conclusion

K-Means successfully segmented customers into meaningful groups. These segments can help banks improve customer relationship management and targeted marketing.

10. References

1. [Kaggle Credit Card Dataset \(CC GENERAL.csv\)](#)
2. [Scikit-learn Documentation](#)
3. [Pandas Documentation](#)
4. Matplotlib Documentation
5. Seaborn Documentation

|Question/Answers|

Q1. Differentiate between Semantic, Instance, and Panoptic Segmentation.

Semantic Segmentation	Instance Segmentation	Panoptic Segmentation
Assigns a class label to every pixel in an image.	Identifies each object instance separately.	Combines semantic and instance segmentation into a single output.
Objects of the same class share the same label.	Each object receives a unique label even if they belong to the same class.	Every pixel has both a class label and an instance ID (for object classes).
Does not distinguish between different objects of the same class.	Distinguishes between different instances of the same class.	Provides complete scene understanding by identifying both objects and background.
	Example: Car 1, Car 2, Car 3 are identified separately.	Example: Road, sky, trees, and each individual car are all identified correct

Q2. What is Fuzzy Logic, and how is it different from Boolean Logic?

Fuzzy Logic is a reasoning approach in which truth values can range between **0 and 1**, allowing partial truth. It is useful for handling uncertainty and real-world situations where decisions are not simply true or false.

Boolean Logic is a traditional logic system where every statement has only two possible values: **True (1)** or **False (0)**.

Difference between Fuzzy Logic and Boolean Logic

Fuzzy Logic

Truth values range between **0 and 1**.

Handles uncertainty effectively.

Allows partial membership.

Used in AI, control systems, washing machines, air conditioners, etc.

Example: Temperature can be **0.7 Hot**.

Boolean Logic

Truth values are only **0 or 1**.

Does not handle uncertainty.

Allows only complete membership.

Used in digital circuits, programming, and databases.

Example: Temperature is either **Hot** or **Not Hot**.

Q3. Suppose I give you a customer dataset containing Age, Annual Income, and Spending Score, but I don't provide any labels.

1. Is this a supervised or unsupervised learning problem?

This is an **Unsupervised Learning** problem because the dataset does not contain any target labels or output classes. The objective is to discover hidden patterns or groups within the data.

2. Which clustering algorithm would you choose first and why?

I would choose **K-Means Clustering** first because:

- It is simple and easy to implement.
- It is computationally efficient for large datasets.
- It works well when clusters are compact and approximately spherical.
- It is widely used for customer segmentation problems.
- It provides meaningful groups based on feature similarity.

3. What assumptions does K-Means make about the data?

K-Means Clustering assumes that:

- The number of clusters (**K**) is known beforehand.
- Clusters are approximately **spherical (round-shaped)**.
- Clusters have similar sizes and densities.

- Features are numerical and preferably standardized.
- Each data point belongs to only one cluster.
- Euclidean distance is an appropriate measure of similarity.